A2.2 Cell structure

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-a2-2.html

Written for this site from the class slides. The chapter is on pages 49–81 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

CELLS AND MICROSCOPES

A2.2.1 Cells: the basic unit of life

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All organisms are built from cells, the basic unit of structure. Using deductive reasoning, cell theory predicts that a newly discovered organism will consist of one or more cells.

A2.2.2 Microscopy skills

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Practise making temporary mounts, staining, focusing with coarse and fine adjustment, and measuring with an eyepiece graticule. **Magnification = image size / actual size.** For example, a cell 8 µm wide drawn 40 mm wide has a magnification of 40 000 / 8 = ×5000.

Watch out: convert to the same units first. 1 mm = 1000 µm.

A2.2.3 Developments in microscopy

pp. 49–81

TECHNIQUE	WHY IT IS USEFUL
ELECTRON MICROSCOPY	Electrons have a shorter wavelength, so resolution is far higher
FLUORESCENT STAINS	Show chosen structures in colour
IMMUNOFLUORESCENCE	Antibodies carry a dye to one specific target
FREEZE FRACTURE	Frozen cells split along the middle of membranes, revealing surfaces
CRYOGENIC ELECTRON MICROSCOPY	Rapid freezing preserves molecules, so that 3D models can be made

TYPES OF CELL

A2.2.4 Structures in all cells

pp. 49–81

All cells have **DNA** as genetic material, **cytoplasm** composed mainly of water, where reactions take place, and a **plasma membrane** of lipids, which separates inside from outside.

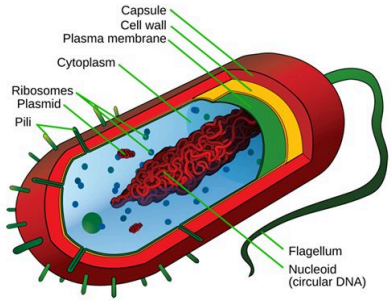
A2.2.5 Prokaryotic cells

pp. 49–81

A Gram-positive eubacterium, such as *Bacillus* or *Staphylococcus*, has a cell wall, plasma membrane, cytoplasm, naked DNA in a loop, and 70S ribosomes.

Prokaryotic cells

A2.2.5



A generalized prokaryote. LadyofHats, public domain.

Prokaryotes (bacteria) have **no nucleus**: their DNA is a **naked loop** in an area called the **nucleoid**.

There are **no membrane-bound organelles**, so the cytoplasm is not divided up.

They have a **cell wall**, small **70S ribosomes**, and often **plasmids**, **pili** and a **flagellum**.

Example to know

Gram-positive bacteria such as Staphylococcus and Bacillus. Be ready to draw one from an electron micrograph.

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A2.2.6 Eukaryotic cells

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A plasma membrane encloses a compartmentalized cytoplasm, with 80S ribosomes. The nucleus has a double membrane with pores, and chromosomes of DNA bound to histones. There is a cytoskeleton of microtubules and microfilaments.

ORGANELLE	FUNCTION
MITOCHONDRION	Aerobic respiration
ROUGH ER	Makes proteins for secretion or membranes
SMOOTH ER	Makes lipids
GOLGI APPARATUS	Modifies and packages proteins
LYSOSOME	Digests food and old organelles

LIFE AND VARIETY

A2.2.7 Life in a single cell

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Single-celled organisms carry out **homeostasis, metabolism, nutrition, movement, excretion, growth, response to stimuli and reproduction**. In *Paramecium*, for example, contractile vacuoles expel water, food vacuoles digest food, and cilia provide movement.

A2.2.8 Animals, fungi and plants

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	ANIMALS	FUNGI	PLANTS
CELL WALL	None	Chitin	Cellulose
VACUOLE	Small, temporary	Large or small	Large, permanent
PLASTIDS	None	None	Chloroplasts and others
CENTRIOLES, CILIA, FLAGELLA	Often present	Mostly none	Mostly none

A2.2.9 Atypical cell structure

pp. 49–81

Many nuclei: aseptate fungal hyphae and skeletal muscle. **No nucleus:** mature red blood cells and phloem sieve tube elements.

SKILLS

A2.2.10 Reading micrographs

pp. 49–81

A **prokaryote** is small, with a nucleoid and a cell wall. A **plant** cell has a cell wall, chloroplasts and a large vacuole. An **animal** cell has a nucleus and mitochondria, but no cell wall. Practise finding each organelle in electron micrographs.

A2.2.11 Drawing and annotating

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Use a sharp pencil and clear, continuous lines, without shading. Keep to the proportions of the micrograph. Annotate each structure with a function: for example, microvilli increase the surface area for absorption.

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